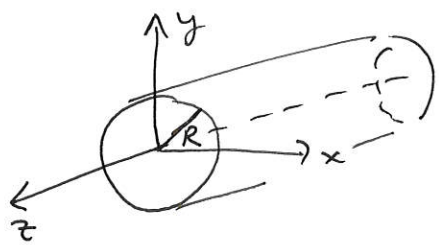
Laplace equation via separation of variables

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0 \quad \rightarrow \text{Example 3.3 of Griffiths}$$

Spherical
$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial \phi}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial \phi}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 \phi}{\partial \varphi^2} = 0$$

Cylindrical
$$\frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial \phi}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 \phi}{\partial \varphi^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$$



Pipe is running along $\hat{z}$.

Problem Metal pipe of radius R, infinitely long and grounded, in ~~the~~ uniform electric field $\vec{E}(\vec{r}) = E_0 \hat{x}$.

No z-dependence in the problem $\phi(r, \varphi)$; $\frac{\partial \phi}{\partial z} = 0$.

$$\Rightarrow \rho \frac{\partial}{\partial \rho}\left(\rho \frac{\partial \phi}{\partial \rho}\right) + \frac{\partial^2 \phi}{\partial \varphi^2} = 0$$

$$\Rightarrow \phi(\rho, \varphi) = S(\rho)\, Q(\varphi)$$

$\Rightarrow$ plug in and divide by SQ:

$$\underbrace{\frac{\rho}{S(\rho)} \frac{\partial}{\partial \rho}\left(\rho \frac{\partial S(\rho)}{\partial \rho}\right)}_{C_1} + \underbrace{\frac{1}{Q}\frac{\partial^2 Q}{\partial \varphi^2}}_{C_2} = 0$$

$C_1 + C_2 = 0 \Rightarrow$ choose $C_1 = \nu^2$

$C_2 = -\nu^2$

$\Rightarrow$ two equations:

$$\begin{cases} \dfrac{d^2 Q}{d\varphi^2} = -\nu^2 Q & (1) \\ \rho \dfrac{d}{d\rho}\left(\rho \dfrac{dS}{d\rho}\right) = \nu^2 S(\rho). & (2) \end{cases}$$

$\Rightarrow \quad Q(\varphi) = A_\nu \cos(\nu\varphi) + B_\nu \sin(\nu\varphi) \quad$ for $\nu \neq 0$

(For $\underline{\nu = 0}$ $\quad Q(\varphi) = A_0 + B_0 \varphi$.)

Try solving (2) with $S(\rho) = a\rho^\alpha$

$$\rho \frac{d}{d\rho}\left(\rho \frac{d\, a\rho^\alpha}{d\rho}\right) = \nu^2 a\rho^\alpha$$

$$a \frac{d}{d\rho}\left(\alpha \rho \rho^{\alpha-1}\right) = \nu^2 a \rho^{\alpha-1}$$

$$\alpha^2 \rho^{\alpha-1} = \nu^2 \rho^{\alpha-1} \Rightarrow \alpha = \pm\nu.$$

$$\Rightarrow S(\rho) = a_\nu \rho^\nu + b_\nu \rho^{-\nu} \quad \text{for } \nu \neq 0.$$

(For $\underline{\nu = 0}$ special case: $\rho \dfrac{d}{d\rho}\left(\rho \dfrac{dS}{d\rho}\right) = 0$

$$\Rightarrow \rho \frac{dS}{d\rho} = \text{const}$$

$$\Rightarrow S(\rho) = a_0 + b_0 \ln\left(\frac{\rho}{\rho_0}\right).)$$

General solution of 2d Laplace with cylindrical symmetry (and no z-dependence)

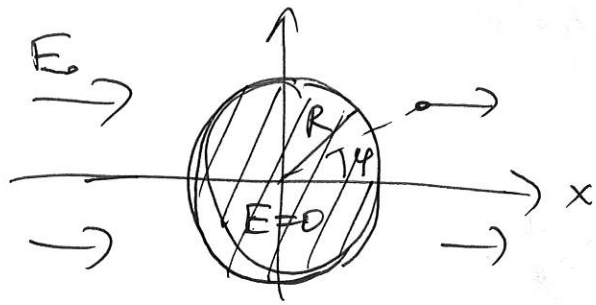
$$* \quad \Phi(\rho, \varphi) = \left[a_0 + b_0 \ln\left(\frac{\rho}{\rho_0}\right)\right](A_0 + B_0\varphi) + \sum_{\nu=1}^{\infty}\left(a_\nu \rho^\nu + b_\nu \rho^{-\nu}\right) \cdot$$

$$\cdot \left[A_\nu \cos(\nu\varphi) + B_\nu \sin(\nu\varphi)\right]$$

<u>Example 2</u> Our initial problem:

Boundary conditions: $\phi(R, \varphi) = 0$ (surface of cylinder)

$$\phi(\rho, \varphi) \to -E_0 x$$



$x = \rho \cos\varphi$

$$\Rightarrow \phi(\rho, \varphi) \to -E_0 \rho \cos\varphi$$
$$\rho \to \infty.$$

$\Rightarrow$ $b_0 = 0$ so that ($b_0 \ln\left(\frac{\rho}{\rho_0}\right)$ term is excluded).

<u>and</u> $\Rightarrow$ $a_n = 0$ for $n \geq 2$. ~~and $B_n = 0$~~

$$\Rightarrow \phi(\rho, \varphi) = a_0 + a_1 \rho \left(A_1 \cos\varphi + \underset{\overset{\parallel}{0}}{B_1} \sin\varphi\right)$$
$$+ \sum_{n=1}^{\infty} b_n \rho^{-n} \left(A_n \cos(n\varphi) + B_n \sin(n\varphi)\right);$$

Clearly $a_1 A_1 = -E_0$.

On the surface of cylinder $\phi(\rho = R, \varphi) = 0$

$$\Rightarrow a_0 = 0$$

$$-E_0 R \cos\varphi + \sum_{n=1}^{\infty} \frac{b_n}{R^n} \left(A_n \cos(n\varphi) + B_n \sin(n\varphi)\right) = 0$$

$\Rightarrow B_n = 0$, $A_n = 0$ for $n \geq 2$.

$$b_1 A_1 / R = +E_0 R \Rightarrow b_1 A_1 = +E_0 R^2$$

$$\Rightarrow \phi(\rho,\varphi) = -E_0 \rho \cos\varphi + \frac{E_0 R^2}{\rho}\cos\varphi =$$

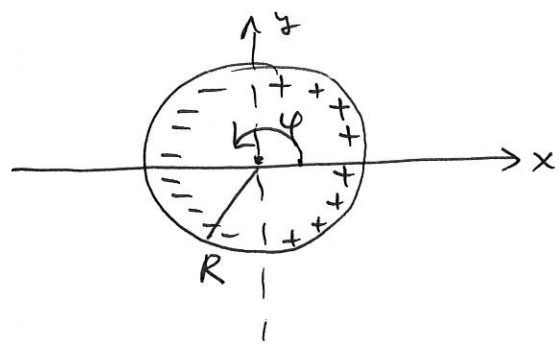
$$= -E_0\left(\rho - \frac{R^2}{\rho}\right)\cos\varphi.$$

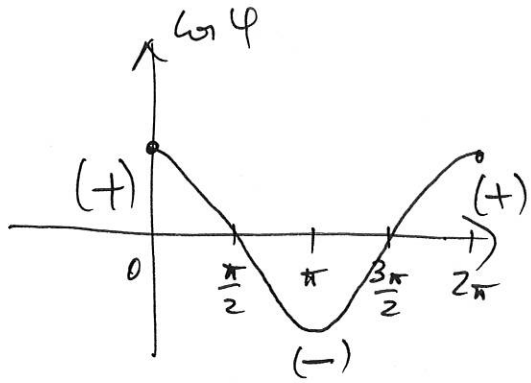
Induced surface charge density

$$\frac{\sigma}{\varepsilon_0} = E_\rho(\rho=R+0) - \underbrace{E_\rho(\rho=R-0)}_{\phi} = -\frac{\partial\phi}{\partial\rho}\Big|_{\rho=R} =$$

$$= E_0\left(1+\frac{R^2}{\rho^2}\right)\cos\varphi\Big|_{\rho=R} = 2E_0\cos\varphi.$$

<u>Find induced dipole moment</u>



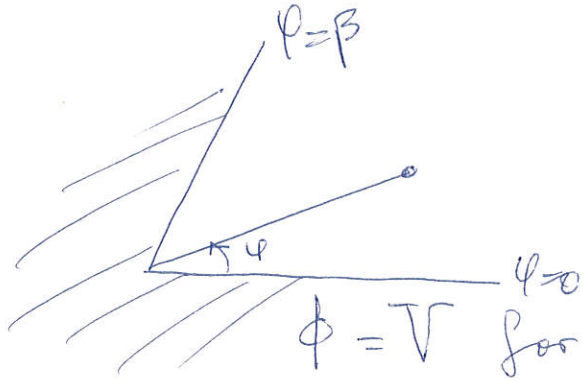


Total charge $Q = 2\varepsilon_0 E_0 \int_0^{2\pi} \cos\varphi \, d\varphi = 0.$

(3.98) $\vec{p} = \int \vec{r}\,' \rho(\vec{r}\,')\, dV' \rightarrow \int \vec{r}\,' \sigma(\vec{r}\,')\, dS'$

because $\rho(r'<R)=0.$ integral over the surface

Conducting Wedge



$\phi = V$ for $\varphi=0$ and $\varphi=\beta$ for all $\rho \geq 0$

$$\Rightarrow \begin{cases} b_0 = 0, \; B_0 = 0 \\ A_\nu = 0, \; b_\nu = 0 \end{cases}$$ (so that $\rho \to 0$ limit is regular)

$$\phi(\rho,\varphi) = a_0 + \sum_{\nu=1}^{\infty} a_\nu \rho^\nu B_\nu \sin(\nu\varphi)$$

and $\sin(\nu\beta) = 0 \Rightarrow \nu\beta = \pi m$

$$\nu = \frac{\pi m}{\beta}$$

$$a_0 = V$$

$$\phi = V + \sum_{m=1}^{\infty} a_m \rho^{\pi m/\beta} \sin\left(\frac{\pi m}{\beta}\varphi\right)$$

$\hookrightarrow$ determined by large-ρ conditions

For $\rho \to 0$: $\rho^{\frac{\pi}{\beta}}$ is most important.

$$\left.\begin{aligned} E_\rho &= -\frac{\partial\phi}{\partial\rho} \sim \rho^{\frac{\pi}{\beta}-1} \\ E_\varphi &= -\frac{1}{\rho}\frac{\partial\phi}{\partial\varphi} \sim \rho^{\frac{\pi}{\beta}-1} \end{aligned}\right\}$$ singular for $\frac{\pi}{\beta} - 1 < 0 \Rightarrow \beta > \pi$

$$\sigma(\rho) = \varepsilon_0 E_\varphi(\rho, \varphi=0) = -\varepsilon_0 \frac{\pi a_1}{\beta} \rho^{\frac{\pi}{\beta}-1}$$

at $\varphi = 0, \beta$

Spherical coordinates

$$\frac{1}{r}\frac{\partial^2}{\partial r^2}(r\phi) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\phi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\phi}{\partial\varphi^2} = 0$$

let $\phi(r,\theta,\varphi) = \frac{U(r)}{r} P(\theta)\Phi(\varphi)$

Plug in and multiply by $r^2\sin^2\theta$ $\left[\frac{1}{U}\frac{d^2U}{dr^2} + \frac{1}{Pr^2\sin\theta}\frac{d}{d\theta}\sin\theta\frac{dP}{d\theta}\right.$

$$+ \underbrace{\frac{1}{Q}\frac{d^2Q}{d\varphi^2}}_{-m^2} = 0$$

$$\Rightarrow \frac{d^2Q}{d\varphi^2} = -m^2 Q, \quad Q = e^{\pm i m\varphi}$$

$$\frac{r^2\sin^2\theta}{U}\frac{d^2U}{dr^2} + \frac{\sin\theta}{P}\frac{d}{d\theta}\left(\sin\theta\frac{dP}{d\theta}\right) - m^2 = 0$$

$$\underbrace{\frac{r^2}{U}\frac{d^2U}{dr^2}}_{\text{function of } r} + \underbrace{\frac{1}{P\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dP}{d\theta}\right) - \frac{m^2}{\sin^2\theta}}_{\text{function of } \theta} = 0$$

$$\text{function of } r = \ell(\ell+1)$$

where ℓ is (complex) constant

$$\frac{d^2U}{dr^2} - \frac{\ell(\ell+1)}{r^2}U = 0 \quad \text{(radial eqn)}$$

$\rightarrow$ try $U = r^\alpha \Rightarrow \alpha = -\ell$, $\alpha = \ell+1$

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dP}{d\theta}\right) + \left(\ell(\ell+1) - \frac{m^2}{\sin^2\theta}\right)P = 0 \quad \text{Legendre eqn}$$

let $x = \cos\theta \Rightarrow$

$$\frac{d}{dx}\left((1-x^2)\frac{dP}{dx}\right) + \left(\ell(\ell+1) - \frac{m^2}{1-x^2}\right)P = 0$$ generalized Legendre eqn.

solved by associated Legendre functions

If $m=0$ so that Φ is <u>azimuthally invariant</u>

$$\Rightarrow \frac{d}{dx}\left((1-x^2)\frac{dP}{dx}\right) + \ell(\ell+1)P = 0$$ (ordinary Legendre eqn)

solution by power series

$$\begin{cases} P_0(x) = 1 \\ P_1(x) = x \\ P_2 = \frac{1}{2}(3x^2-1) \\ P_3 = \frac{1}{2}(5x^3-3x) \end{cases}$$

$$\Rightarrow \phi(r,\theta) = \sum_{\ell=0}^{\infty}\left[A_\ell r^\ell + \frac{B_\ell}{r^{\ell+1}}\right]P_\ell(\cos\theta)$$